Unit 3 Measurement Test

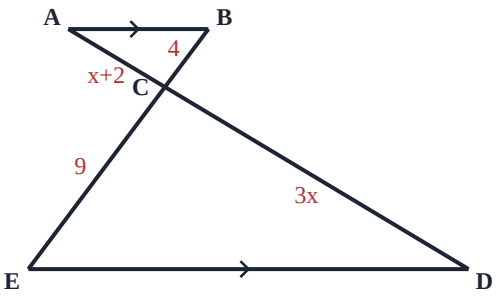
Math 9E Lessons 3.1–3.3 8 questions · 30 marks Name: _____

Show all your work. Round to the nearest tenth unless a question says otherwise. Answers are on the last page.

Question 1

[3 marks]

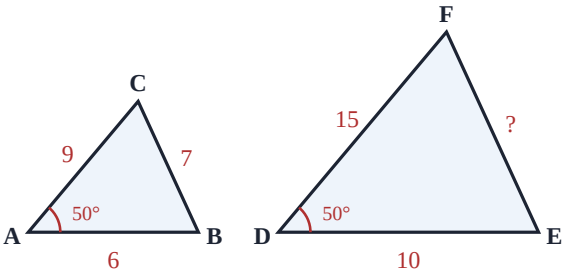
In an hourglass figure, AD and BE cross at C and AB ∥ ED. AC = x + 2, CD = 3x, BC = 4 and CE = 9. Find x, AC and CD.



Question 2

[4 marks]

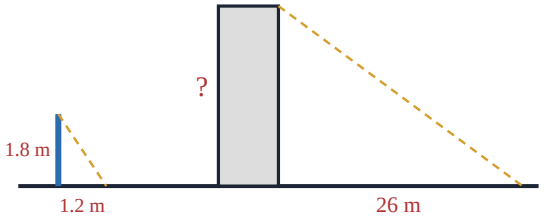
In ΔABC, AB = 6, AC = 9 and ∠A = 50°. In ΔDEF, DE = 10, DF = 15 and ∠D = 50°. (a) Prove the triangles are similar. (b) If BC = 7, find EF.



Question 3

[3 marks]

A 1.8 m pole casts a 1.2 m shadow. At the same time a building casts a 26 m shadow. (a) Find the building's height. (b) Later, the building's shadow is 39 m. How long is the pole's shadow then?



Question 4

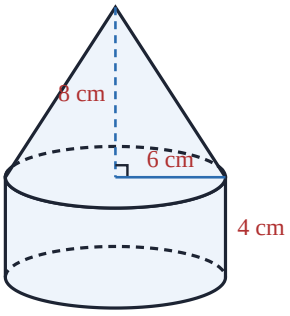
[3 marks]

(a) A doctor orders 2 g of Ampicillin per day, in 4 equal doses. The capsules are 250 mg. How many capsules per dose and per day? (b) 1 L of saline must drip in over 8 hours. What is the rate in mL per minute?

Question 5

[4 marks]

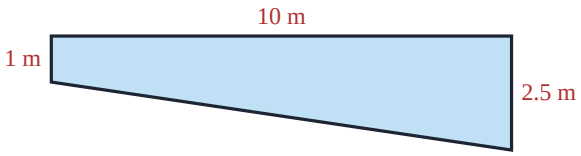
A cone sits on top of a cylinder, both with radius 6 cm. The cylinder is 4 cm tall and the cone is 8 cm tall (height, not slant). Find the total outside surface area, including the bottom.



Question 6

[5 marks]

A pool is 10 m long and 4 m wide. The depth slopes from 1 m to 2.5 m. Tile the bottom, both side walls and both end walls. (a) Find the area to be tiled. (b) Tiles cost \$40 USD/m^2 and 1 USD = 1.38 CAD. Find the cost in CAD.

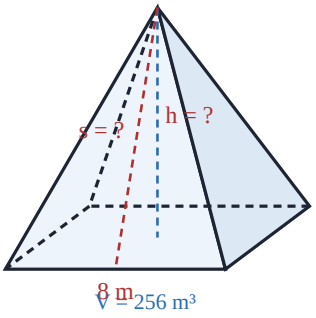


side view · width 4 m

Question 7

[4 marks]

A right square pyramid has volume 256 m^3 and base side 8 m. Find its height and its slant height to the nearest hundredth.

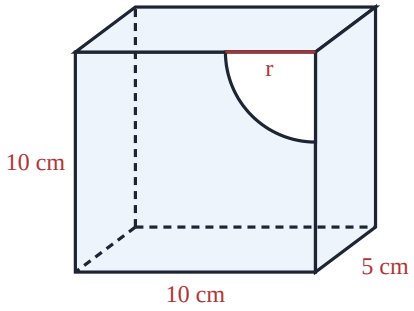


$V = 256\ m^3$

Question 8

[4 marks]

A quarter cylinder is cut from one edge of a 10 cm × 10 cm × 5 cm cuboid. The quarter cylinder runs the 5 cm length. The remaining volume is 400 cm^3. Find the radius.



ANSWER KEY

Q1 $\Delta ABC \sim \Delta DEC \Rightarrow AC/DC = BC/EC$: $(x + 2)/(3x) = 4/9 \Rightarrow 9x + 18 = 12x \Rightarrow$ **$x = 6$, $AC = 8$, $CD = 18$.**

Q2 (a) $DE/AB = 10/6 = 5/3$ and $DF/AC = 15/9 = 5/3$, and the included angles are equal ($\angle A = \angle D = 50°$) $\Rightarrow$ **$\Delta ABC \sim \Delta DEF$ (SAS).** (b) $EF = 7 \times 5/3 \approx$ **11.67**.

Q3 (a) $H/26 = 1.8/1.2 \Rightarrow$ **$H = 39$ m**. (b) Now height = shadow (39 m : 39 m), so the pole's shadow = **1.8 m**.

Q4 (a) 2 g = 2000 mg ÷ 4 = 500 mg per dose $\Rightarrow$ **2 capsules per dose, 8 per day**. (b) 1000 mL ÷ 480 min ≈ **2.08 mL/min**.

Q5 Slant = $\sqrt{(6^2 + 8^2)} = 10$. Cone $\pi(6)(10) = 60\pi$; cylinder side $2\pi(6)(4) = 48\pi$; base 36π. Total = **$144\pi \approx 452.4$ cm²**.

Q6 (a) Sloped bottom: length $\sqrt{(10^2 + 1.5^2)} \approx 10.11$ m × 4 ≈ 40.45 m². Two trapezoid sides: $2 \times (1 + 2.5)/2 \times 10 = 35$ m². Ends: $1 \times 4 + 2.5 \times 4 = 14$ m². Total ≈ **89.45 m²**. (b) $89.45 \times 40 \times 1.38 \approx$ **$4937.50 CAD**.

Q7 $\frac{1}{3}(64)h = 256 \Rightarrow$ **$h = 12$ m**. Slant = $\sqrt{(12^2 + 4^2)} = \sqrt{160} \approx$ **12.65 m**.

Q8 Cuboid = 500 $\Rightarrow$ removed = $100 = \frac{1}{4}\pi r^2(5) \Rightarrow r^2 = 80/\pi \approx 25.46 \Rightarrow$ **$r \approx 5.05$ cm**.